

RUBEN DARIO FLOREZ ZELA

Electronic Engineer · Human-Centered AI for Intelligent Vehicles · Computer Vision & Embedded AI

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ORCID | [Google Scholar](#) | [LinkedIn](#) | [GitHub](#) | [Personal Web](#)

RENACYT Level V (CONCYTEC) · Google Scholar: 217 citations · h-index 5 · i10-index 3 (2026)

RESEARCH PROFILE

Electronic engineer and AI researcher working at the intersection of machine learning, embedded systems, automated driving, and safety-critical intelligent vehicles. My research includes conformal risk control for SAE Level 2 automated driving, deployment-aware evaluation of vision models, and real-time AI on embedded hardware. I am interested in advancing human-centered perception and decision-making for intelligent vehicles, combining driver monitoring, computer vision, embedded AI, and safety-aware machine learning for ADAS and automated driving.

EDUCATION

M.Sc. in Electronic Engineering (Automation & Instrumentation)

Expected 2026

Universidad Nacional de San Agustín de Arequipa (UNSA), Arequipa, Peru

- **Thesis:** *Multi-Horizon Prediction of Delayed-Response Risk and Three-State Conformal Supervision for Longitudinal Fallback in SAE Level 2 Automated Driving.*
- **Relevant coursework:** Linear and Nonlinear Systems, Signal Processing, Industrial Instrumentation, Digital Control, Real-Time Automation, Intelligent Control, and Special Topics in Automation.

Bachelor's Degree in Electronic Engineering

Degree awarded Feb. 2023

Universidad Nacional de San Antonio Abad del Cusco (UNSAAC), Cusco, Peru

- 223 approved academic credits; relevant coursework: Artificial Intelligence, Robotics, Microprocessors and Computer Architecture, Digital Signal Processing, Control Theory, Industrial Control Systems, Instrumentation, and Telecommunications.

Professional Title: Electronic Engineer

Awarded Feb. 2024

UNSAAC, Cusco, Peru | Professional License (CIP): 336757 | Obtained by thesis

- **Professional thesis:** Design and Implementation of a Real-Time Driver Drowsiness Detection System Using Computer Vision and Convolutional Neural Networks, **unanimously approved with excellence (19/20).**

SELECTED RESEARCH EXPERIENCE

Safety-Critical Risk Control for Automated Driving (M.Sc. Research)

2025 – 2026

Universidad Nacional de San Agustín de Arequipa (UNSA)

- Developed driver-grouped machine-learning models for multi-horizon prediction of delayed-response risk in SAE Level 2 automated driving.
- Designed calibrated and conformal risk-control mechanisms for graded longitudinal intervention with HUMAN-ASSIST-FALLBACK supervision.
- Evaluated safety effects through counterfactual longitudinal simulation under driver-specific uncertainty.

Deployment-Aware Vision Benchmarking for Driver Monitoring (Lead Researcher)

2026 – Present

Independent research / public reproducibility repository

- Benchmarking lightweight CNN and transformer-based vision models under clean, corrupted, and cross-domain conditions.
- Evaluating embedded deployment using NVIDIA Jetson Nano and TensorRT with latency, throughput, memory, and reliability criteria.
- Maintaining frozen model semantics, reproducible protocols, manifests, and audit-oriented evaluation pipelines.

Embedded Vision for Driver-State Monitoring (Lead Researcher)

2023

LIECAR Lab, UNSAAC

- Developed CNN-based real-time driver monitoring using eye and facial-region analysis.
- Designed ROI correction procedures robust to illumination variability.
- Deployed real-time inference on NVIDIA Jetson Nano.

Robotic Control Systems (Research Collaborator)

2023

LIECAR Lab (UNSAAC) & PAVIC Lab (UFAC, Brazil)

- Developed PID and fuzzy-logic controllers for a tendon-driven 3D-printed robotic lumbar-spine mechanism.

SELECTED RESEARCH OUTPUTS

Current manuscripts and preprints

- [1] **Florez, R. D.** *Driver-Grouped Conformal Risk Control for Graded Longitudinal Intervention in Level 2 Automated Driving: A Counterfactual Simulation Study*. Manuscript submitted to *IEEE Transactions on Intelligent Vehicles*, 2026.
- [2] **Florez, R. D.** *Beyond Aggregate Scores: Deployment-Aware and Non-Compensatory Benchmarking of Vision-Based Eye-State Recognition Models for Driver Monitoring*. arXiv preprint (2606.08123), 2026.

Peer-reviewed journal articles

- [1] **Florez, R.**, Palomino-Quispe, F., Alvarez, A. B., Coaquira-Castillo, R. J., & Herrera-Levano, J. C. (2024). A Real-Time Embedded System for Driver Drowsiness Detection Based on Visual Analysis of the Eyes and Mouth Using Convolutional Neural Network and Mouth Aspect Ratio. *Sensors*, 24(19), 6261.
- [2] **Florez, R.**, Palomino-Quispe, F., Coaquira-Castillo, R. J., Herrera-Levano, J. C., Paixão, T., & Alvarez, A. B. (2023). A CNN-Based Approach for Driver Drowsiness Detection by Real-Time Eye State Identification. *Applied Sciences*, 13(13), 7849.

Selected peer-reviewed conference paper

- [1] Paixão, T., Alvarez, A. B., **Florez, R.**, & Palomino-Quispe, F. (2023). Fuzzy Control for Simplified Lumbar Spine Robotic Mechanism Motion. *2023 IEEE International Conference on Networking, Sensing and Control (ICNSC)*, pp. 1–6. IEEE.

Full publication list: [Google Scholar](#)

ACADEMIC POSITIONS

Contract Lecturer – Electronic Engineering Department

Aug. 2024 – Jan. 2026

Universidad Nacional de San Antonio Abad del Cusco (UNSAAC), Cusco, Peru

- Taught Artificial Intelligence, Digital Image Processing, Robotics, and Electronics Laboratories; supervised applied student work in machine learning, computer vision, and embedded systems.

Teaching Assistant – Electronic Engineering Department

Apr. 2024 – Aug. 2024

Universidad Nacional de San Antonio Abad del Cusco (UNSAAC), Cusco, Peru

- Supported practical and laboratory sessions in electronics and AI-related courses.

TECHNICAL SKILLS

Programming:	Python, C/C++, MATLAB
Machine Learning:	PyTorch, TensorFlow, Keras, scikit-learn; classification, calibration, model evaluation
Computer Vision:	OpenCV, RGB/NIR imaging, gaze analysis, facial landmark tracking, image classification
Embedded & Edge AI:	NVIDIA Jetson Nano, TensorRT, CUDA-enabled inference, Raspberry Pi, STM32, ESP32
Control & Real-Time Systems:	Digital control, PID/fuzzy control, real-time automation, instrumentation
Research Software:	Linux, Git/GitHub, reproducible experiment organization, deterministic evaluation workflows

RESEARCH INTERESTS

· Human-Centered AI · Driver Monitoring · Intelligent Vehicles · Automated Driving · Computer Vision · Predictive Perception · Embedded & Edge AI · Safety-Critical Machine Learning · Intelligent Transportation Systems

SELECTED DISTINCTIONS, SERVICE & LANGUAGES

- **RENACYT Level V Researcher**, CONCYTEC, Peru; IEEE Member (ITSS and VTS).
- **1st Place:** VIII Regional Science, Technology & Innovation Fair (2023) and II International Congress of Research, Innovation & Entrepreneurship, UNSAAC (2022).
- Reviewer for *Scientific Reports*, *Scientific Data*, and *ACM Transactions on Intelligent Systems and Technology*.
- **Languages:** Spanish (native); English (intermediate, working proficiency for academic research and scientific communication).